

EDUCATION**Nanyang Technological University, Singapore****January 2025 - present***Ph.D.* candidate, Medicine (GPA: 4.88/5.0)

Advisors: Asst. Prof. Jue Tao Lim, Dr. Ian Wee Liang En (Singapore General Hospital)

- Research interests: Systems Modelling, Antimicrobial Resistance, Sexually Transmitted Infections.
- [[Google Scholar](#)].

National University of Singapore, Singapore**August 2019 - June 2023***B.Eng. (Hons)*, Computer Engineering (Distinction, GPA: 4.35/5.0)

Advisors: Dr. Ganesh Neelakanta Iyer, Dr. Ravi Suppiah, Asst. Prof. Deepu John (University College Dublin)

- Research interests: Deep Learning for Engineering.
- University Town College Program.

PUBLICATIONS

1. **Zihao Wang**, Dariya Nikitin, George T.B. Young, Matan Yechezkel, Liang En Wee, Martin T.W. Chio, Lin Geng, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Joseph A. Lewnard, Lilith K. Whittles, and Jue Tao Lim, “The long-term epidemiological impacts of doxycycline post-exposure prophylaxis and vaccination against multidrug-resistant *Neisseria gonorrhoeae*: a mathematical modelling study,” under review by *Science Translational Medicine*. [[medRxiv](#), [GitHub](#)].
 2. Lin Geng*, Perran A. Ross*, Cai Yu*, Tao Huang, Jo Yi Chow, **Zihao Wang**, Esther Li Wen Choo, Chia-Chen Chang, Lisa Couper, Xinyue Gu, Ary Hoffmann, and Jue Tao Lim, “Robustness of Wolbachia-mediated incompatible-insect technique to future climate change scenarios,” submitted. [[medRxiv](#)].
 3. An Ting Tay*, Peihong Guo*, Jinghao Nicholas Ngaim*, **Zihao Wang***, Liang En Wee, Calvin Chiew, Lalitha Kurupatham, Lee Ching Ng, Po Ying Chia, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim*, “Risk of post-acute multi-organ sequelae after dengue reinfection,” submitted.
 4. Liang En Wee*, Reen Wan Li Ho*, **Zihao Wang***, Peihong Guo*, Rhea Khanna, Calvin Chiew, Russell Jingxian Li, Jiahui Li, Chee-Fu Yung, Chia Yin Chong, Zubaidah Said, David Chien Boon Lye, Vernon Lee, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization following paediatric hospitalization for respiratory-syncytial-virus versus influenza or Omicron COVID-19 in a tropical setting: a population-based cohort study,” submitted.
 5. Peihong Guo*, Reen Wan Li Ho*, **Zihao Wang***, Ian Liang En Wee*, Russ Li, David Chien Boon Lye, Kelvin Bryan Tan, Jue Tao Lim, “Acute and post-acute healthcare utilization after hospitalization for COVID-19, influenza or respiratory syncytial virus in adults: a population-based cohort study,” submitted.
 6. **Zihao Wang**, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, “Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission,” *Nature Health* 1(7), 724-736, 2026. [[Paper](#), [GitHub](#)].
 7. **Zihao Wang**, Yuhao Li, Yao Shi, and Zhe Wu, “Input convex lipschitz recurrent neural networks for robust and efficient process modeling and optimization,” arXiv. [[arXiv](#), [GitHub](#)].
 8. **Zihao Wang**, and Zhe Wu, “Towards foundation model for chemical reactor modeling: Meta-learning with physics-informed adaptation,” *Chemical Engineering Research and Design* 218, 839-853, 2025. IF: 4.0, JCR Q2. [[Paper](#), [GitHub](#)].
 9. **Zihao Wang**, Donghan Yu, and Zhe Wu, “Real-time machine-learning-based optimization using input convex long short-term memory network,” *Applied Energy* 377, 124472, 2025. IF: 12.2, JCR Q1. [[Paper](#), [GitHub](#)].
 10. **Zihao Wang**, and Ravi Suppiah, “Upper limb movement recognition utilising EEG and EMG signals for rehabilitative robotics,” *Future of Information and Communication Conference*, 676-695, 2023. [[Paper](#), [GitHub](#)].
- (* denotes equal contribution)

PRESENTATIONS

1. **Zihao Wang***, Dariya Nikitin, George T.B. Young, Matan Yechezkel, Liang En Wee, Martin T.W. Chio, Lin Geng, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Joseph A. Lewnard, Lilith K. Whittles, and Jue Tao Lim, "The long-term epidemiological and evolutionary consequences of joint doxycycline post-exposure prophylaxis and vaccination deployment against multidrug-resistant *Neisseria gonorrhoeae*: a mathematical modelling study," *20th Singapore Public Health & Occupational Medicine (PHOM) Conference*, Singapore, 2026. (**Oral, Award**).
2. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission," *Singapore HIV, Hepatitis and STI Congress (SHHSC)*, Singapore, 2026. (**Oral**).
3. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Modelling the interplay between doxycycline post-exposure prophylaxis, four-component serogroup B meningococcal (4CMenB) vaccine, and antimicrobial resistance in *Neisseria gonorrhoeae*," *European Congress of Clinical Microbiology and Infectious Diseases (ESCMID Global 2026)*, Munich, Germany, 2026. (Poster, **Top 2%**).
4. **Zihao Wang***, Dariya Nikitin, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Modelling the interplay between doxycycline post-exposure prophylaxis, four-component meningococcal serogroup B (4CMenB) vaccine, and antimicrobial resistance in *Neisseria gonorrhoeae*," *LKCMedicine Early Researcher Network (LEARN) Symposium*, Singapore, 2026. (**Flash Talk**).
5. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-run public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Epidemics: 10th International Conference on Infectious Disease Dynamics (Epidemics10)*, San Diego, California, USA, 2025. (**Oral**).
6. **Zihao Wang***, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Singapore Health & Biomedical Congress (SHBC)*, Singapore, 2025. (Poster).
7. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Accelerated modeling of various chemical processes using meta-learning-based foundation models: A few-shot learning approach using Reptile," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (**Oral**). [[Abstract](#)].
8. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Input convex LSTM for fast machine learning-based optimization," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (Poster). [[Abstract](#)].
9. **Zihao Wang**, and Zhe Wu*, "Input convex LSTM: A convex approach for fast model predictive control," Institute for Machine Learning (Head: Professor Sepp Hochreiter), Johannes Kepler University, Linz, Austria, December 2023. (**Invited talk** by Professor Günter Klambauer).

(* denotes presenting author)

AWARDS AND HONORS

Best Oral Presentation , Resident & Student Category, 20th PHOM conference	08/2026
Travel Grant , ESCMID Global 2026	04/2026
Top Rated Abstract , ESCMID Global 2026	04/2026
NTU Research Scholarship , NTU	2025 - Present
IEEE Singapore Computer Society Prize , NUS	06/2023
Outstanding Computing Project Prize , NUS	06/2023

DISSERTATIONS

1. **Zihao Wang**, “Multimodal sensor fusion using AI for healthcare IoT sensors,” B.Eng. Dissertation, Computer Engineering, National University of Singapore, Singapore, 2023. [[Paper](#)].
2. **Zihao Wang**, “AI research for rehabilitative robotics,” Undergraduate Research Opportunity Project (UROP) Thesis, Computer Engineering, National University of Singapore, Singapore, 2022. [[Paper](#)].

PROFESSIONAL EXPERIENCES

National Centre for Infectious Diseases

May 2025 - May 2026

Research Staff (Part-time), Tan Tock Seng Hospital, Singapore

Advisors: Adjunct Assoc. Prof. Kelvin Bryan Tan, Asst. Prof. Jue Tao Lim

- Estimation of infection fatality ratio (IFR) during pandemics, population-based retrospective cohort studies, respiratory syncytial virus (RSV), dengue

Department of Chemical & Biomolecular Engineering

July 2023 - January 2025

Research Engineer (Full-time), National University of Singapore, Singapore

Advisor: Asst. Prof. Zhe Wu

- Input convex neural networks, model predictive control, chemical process modeling, meta-learning, physics-informed learning

Department of Biomedical Engineering

March 2023 - July 2023

Undergraduate Research Assistant (Internship), National University of Singapore, Singapore

Advisor: Asst. Prof. Yueming Jin

- Computer vision, gastrectomy surgical video analysis, modality-cataract video segmentation

Rolls-Royce@NTU Corporate Laboratory

May 2022 - August 2022

Research Engineer (Internship), Nanyang Technological University, Singapore

Advisors: Dr. Aditya Venkataraman, Dr. Benjamin Cheong

- Software engineering, hybrid-electric power and propulsion design tools, marine and aero engines

School of Computing

March 2022 - May 2022

Undergraduate Research Assistant (Internship), National University of Singapore, Singapore

Advisor: Prof. Shengdong Zhao

- Human-computer interaction

SKILLS

Computer: C, C++, Java, Python, R, Assembly Language, MATLAB, PostgreSQL, Verilog, Vensim, Gephi, LaTeX

Languages: Mandarin (native), English (bilingual), Cantonese (proficient), Korean (conversational)